MADA: A Multi-Agent AI System for Autonomous Database Administration

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Abstract

Database administration — indexing, configuration, query tuning, incident diagnosis and security monitoring — is still largely manual, and the AI tools that automate it each cover one duty and act on predictions rather than measurements. This project proposes MADA (Multi-Agent Database Administrator), a multi-agent system for autonomous PostgreSQL administration. Six specialist agents (monitor, index, knob, query, diagnosis and security) read shared telemetry, test candidate actions on a sandbox clone of the database, and submit structured proposals to a coordinator that resolves conflicts over shared resources and to a safety gate that admits only changes with measured gain, keeps irreversible actions for human approval and rolls back regressions. A review of twenty-two published systems shows five gaps that MADA addresses: single-duty tools, trust in estimates over measurements, no safety layer for agent actions, security handled apart from performance, and evaluation in isolation. A working prototype has been built on PostgreSQL 16 with a 168 MB e-commerce database and a live eight-session workload; in a first tuning run it cut the 95th-percentile latency of the slowest report from 20.1 s to 39 ms.

Introduction

Almost every modern application depends on a relational database, and that database stays fast and safe only through continuous expert work: choosing indexes, setting memory and planner parameters, rewriting slow SQL, diagnosing incidents and watching for misuse. Pavlo et al. argued that a database management system should take on much of this work itself — a “self-driving” system that forecasts its own workload and plans its own changes [1]. In the years since, research on AI for databases has produced learned components for nearly every one of these duties [2]: knob tuners that learn configurations from earlier tuning sessions [3], cloud services that index millions of databases automatically [4], learned optimisers that steer query plans with hints [6], and diagnosis tools that explain anomalies from performance metrics [7].

These advances arrive as separate tools, each owning one duty and each trusting its own predictions, and two problems follow. First, the duties interact. A new index changes query plans, memory use and the cost of every write, so a knob tuner, an index advisor and a query rewriter running side by side can undo one another's work. Second, a prediction is not an outcome: Ding et al. showed that optimiser cost estimates frequently recommend indexes that make real queries slower [5]. Security sits further apart still — detectors that profile each role's normal access [9] usually run outside the tuning loop, so nothing decides whether a performance change or a security response should come first.

Large language models add a new option and a new risk. An LLM agent can read documentation and diagnose incidents [8], and several agents can divide a task between them through conversation [10]. But an agent that holds database credentials can also run a wrong statement directly on production, and research prototypes rarely put a verified safety layer between an agent's proposal and the live system.

This project proposes and builds **MADA (Multi-Agent Database Administrator)**. Six specialist agents — monitor, index, knob, query, diagnosis and security — share one view of a PostgreSQL database, test every idea on a sandbox clone, and send structured proposals to a coordinator and a safety gate. The

coordinator resolves conflicts over shared resources and orders actions by priority (security, availability, correctness, performance). The gate applies only changes backed by measured evidence, keeps irreversible actions for a human, and rolls back any change that makes the system slower. A working prototype has been built on PostgreSQL 16 and runs against a 168 MB e-commerce database under a live eight-session workload. This review presents the literature the design builds on, the gaps it addresses, and the proposed architecture.

Literature Review

Self-driving databases and configuration tuning. The self-driving DBMS vision [1] relies on forecasting what the workload will do next; QB5000 clusters queries into templates and forecasts their arrival rates so that indexes can be built before demand arrives [11]. Configuration tuning has moved from Gaussian-process models that reuse past tuning data (OtterTune [3]) to deep reinforcement learning (CDBTune [12], QTune [13]) and, most recently, to language models that read the manual and narrow the search before Bayesian optimisation (DB-BERT [14], GPTuner [15]). Each of these tunes configuration knobs only, and the learning-based ones need many trial runs on the instance being tuned.

Index selection. Index advisors date back to AutoAdmin's what-if analysis, in which the optimiser costs hypothetical indexes without building them [16]. Later work improved the search — Extend's recursive multi-attribute strategy [17] and deep-reinforcement-learning advisors [19] — and a broad experimental evaluation compared eight algorithms under one framework [18]. Two papers question the cost estimates themselves: Ding et al. show that estimated benefit often disagrees with measured execution [5], and Microsoft's Azure SQL Database validates every automatic index change after deployment and reverts regressions [4].

Query optimisation and rewriting. Neo learned a whole query optimiser from experience [20]; Bao made the approach practical by choosing planner hint sets for an existing optimiser [6]. Query rewriting has been framed as a search over rule applications with Monte Carlo tree search [21] and, with LLM-R2, as an LLM choosing rules from a rule engine so that every rewrite stays valid [22].

Diagnosis and security. DBSherlock explains performance anomalies with predicates that separate the anomaly from normal operation [7]; iSQUAD diagnoses intermittent slow queries in a cloud fleet by matching incidents to known cases [23]; D-Bot uses several collaborating LLMs to reason from documentation and metrics to a root cause [8]. On the security side, Kamra et al. profile which data each role normally touches and flag queries that do not fit [9], and Halfond et al. classify SQL-injection attacks and countermeasures [24].

Agents. ReAct interleaves reasoning with tool calls [25], and AutoGen lets several LLM agents cooperate through conversation [10]. These frameworks supply the agent loop but say nothing about database-specific safety: which statements an agent may run, how its claims are checked, or when a person must approve.

Below are the reference articles that are closest to this topic but differ in implementation.

Table 1. Related work

Ref. No.	Year of publication	Authors and Title	Contributions	Limitations
1	2017	Pavlo et al. Self-Driving Database Management Systems.	1. Architecture for a DBMS that forecasts its workload and plans its own tuning actions. 2. Peloton prototype as a research platform for autonomous operation.	1. Vision paper with an early prototype; no end-to-end evaluation of actions. 2. Tied to a custom in-memory DBMS rather than an existing engine.
3	2017	Van Aken, Pavlo, Gordon and Zhang. Automatic DBMS Tuning Through Large-scale Machine Learning (OtterTune).	1. Reuses data from earlier tuning sessions by mapping a new workload to a known one. 2. Ranks important knobs with Lasso and recommends settings with Gaussian-process regression.	1. Configuration knobs only; no indexes, queries or incidents. 2. Needs a large repository of past observations and many test runs.
11	2018	Ma et al. Query-based Workload Forecasting for Self-Driving DBMSs (QB5000).	1. Clusters queries into templates and forecasts arrival rates per cluster. 2. Uses forecasts to choose indexes ahead of demand.	1. Forecasting only; acting on the forecast is left to other components. 2. Ensemble models add training overhead for small deployments.
12	2019	Zhang et al. An End-to-End Automatic Cloud Database Tuning System Using Deep RL (CDBTune).	1. Deep deterministic policy gradient learns knob settings by trial and error. 2. Reward built from throughput and latency changes; no labelled data needed.	1. Many training iterations, each running the workload. 2. Exploration on a live instance can apply harmful settings.
13	2019	Li, Zhou, Li and Gao. QTune: A Query-Aware Database Tuning System with Deep RL.	1. Featurises SQL queries so tuning reacts to the actual workload. 2. Supports query-, workload- and cluster-level tuning.	1. Knob tuning only. 2. Costly training before useful recommendations.
14	2022	Trummer. DB-BERT: A Database Tuning Tool that Reads the Manual.	1. Extracts tuning hints from manuals and web text with a language model. 2. Uses reinforcement learning to decide which hints to apply.	1. Quality depends on the text available for the DBMS. 2. Configuration only; changes tried directly on the system.
15	2024	Lao et al. GPTuner: Manual-Reading Database Tuning via GPT-Guided Bayesian Optimization.	1. LLM turns manuals and forum text into structured knob knowledge and narrowed ranges. 2. Coarse-to-fine Bayesian optimisation finds good settings in few trials.	1. LLM calls add cost and latency. 2. Covers knobs only; no coordination with indexing or query tuning.

Ref. No.	Year of publication	Authors and Title	Contributions	Limitations
16	1997	Chaudhuri and Narasayya. An Efficient, Cost-Driven Index Selection Tool for Microsoft SQL Server.	1. What-if analysis: the optimiser costs hypothetical indexes without building them. 2. Per-query candidate selection and greedy enumeration for the whole workload.	1. Relies on optimiser cost estimates, which can be wrong. 2. Offline tool; a DBA still decides and applies.
17	2019	Schlosser, Kossmann and Boissier. Efficient Scalable Multi-Attribute Index Selection Using Recursive Strategies (Extend).	1. Recursive greedy extension builds multi-column indexes step by step. 2. Scales to large workloads with good quality under a storage budget.	1. Quality bounded by the cost model. 2. Index selection only.
18	2020	Kossmann, Halfpap, Jankrift and Schlosser. An Experimental Evaluation of Index Selection Algorithms.	1. Compares eight index selection algorithms on TPC-H, TPC-DS and JOB. 2. Open-source evaluation platform with what-if cost support.	1. Evaluation uses estimated costs, not measured run times. 2. No guidance on validating choices before deployment.
19	2020	Lan, Bao and Peng. An Index Advisor Using Deep Reinforcement Learning.	1. Deep Q-network chooses indexes from heuristically generated candidates. 2. Handles multi-attribute indexes and index interactions.	1. Needs training per workload. 2. Reward based on cost estimates.
4	2019	Das et al. Automatically Indexing Millions of Databases in Microsoft Azure SQL Database.	1. Production service that recommends, creates and drops indexes across millions of databases. 2. Validates each change after deployment and reverts regressions.	1. Index management only. 2. Validation happens on production, after the change.
5	2019	Ding et al. AI Meets AI: Leveraging Query Executions to Improve Index Recommendations.	1. Shows optimiser cost estimates often predict index benefit wrongly. 2. Classifier trained on execution data cuts query regressions from index changes.	1. Needs execution history for training. 2. Addresses index tuning only.
20	2019	Marcus et al. Neo: A Learned Query Optimizer.	1. End-to-end learned optimiser using a value network over plan trees. 2. Bootstraps from an existing optimiser and improves with experience.	1. Long training time. 2. Sensitive to schema and workload change.
6	2021	Marcus et al. Bao: Making Learned Query Optimization Practical.	1. Chooses planner hint sets per query with Thompson sampling over a tree model. 2. Runs on top of PostgreSQL with modest training.	1. Limited to the hint sets supplied. 2. Does not rewrite SQL, change indexes or configuration.

Ref. No.	Year of publication	Authors and Title	Contributions	Limitations
21	2021	Zhou, Li, Chai and Feng. A Learned Query Rewrite System using Monte Carlo Tree Search.	1. Searches the order of rewrite-rule application with MCTS. 2. Learned estimator predicts cost reduction of a rewrite.	1. Bounded by the rule set. 2. Relies on estimated rather than measured benefit.
22	2024	Li, Yuan, Wang, Cong and Bing. LLM-R2: An LLM-Enhanced Rule-based Rewrite System.	1. LLM selects rewrite rules for a rule engine, so every rewrite stays valid. 2. Contrastive demonstration selection improves the LLM's choices.	1. LLM inference cost per query. 2. Rewrite only; no link to indexing or monitoring.
7	2016	Yoon, Niu and Mozafari. DBSherlock: A Performance Diagnostic Tool for Transactional Databases.	1. Generates predicates that separate an anomaly region from normal operation. 2. Builds causal models of root causes from user feedback.	1. User must mark the anomaly region. 2. Explains causes but does not apply fixes.
23	2020	Ma et al. Diagnosing Root Causes of Intermittent Slow Queries in Cloud Databases (iSQUAD).	1. Anomaly extraction and dependency cleansing on KPIs for slow queries. 2. Bayesian case-based clustering matches new incidents to known causes.	1. Needs labelled historical cases. 2. Diagnosis only.
8	2024	Zhou et al. D-Bot: Database Diagnosis System using Large Language Models.	1. LLM extracts diagnosis knowledge from documents and reasons over metrics with tree search. 2. Multiple LLMs collaborate to produce a root-cause report.	1. LLM cost, latency and risk of unsupported conclusions. 2. Suggested fixes are not tested before use.
9	2008	Kamra, Terzi and Bertino. Detecting Anomalous Access Patterns in Relational Databases.	1. Learns per-role profiles of SQL access from query logs. 2. Flags queries whose access pattern does not fit the issuing role.	1. Assumes role-based access control and clean training logs. 2. Detection only; response left to administrators.
24	2006	Halfond, Viegas and Orso. A Classification of SQL-Injection Attacks and Countermeasures.	1. Taxonomy of injection mechanisms and attack types. 2. Compares detection and prevention techniques against that taxonomy.	1. Survey; no new detector is evaluated. 2. Predates learned detectors and automated response.

Gaps Identified

- **One duty per tool.** Every system in Table 1 handles a single duty — knobs, indexes, queries, diagnosis or security. None coordinates them, although an index changes plans, memory and write cost at once.
- **Predictions trusted over measurements.** Most tuners act on optimiser cost estimates or model predictions. Execution-based validation [4], [5] exists for indexes only, and in [4] it happens after the change is already live.

- **No safety layer for agent actions.** LLM-based tools [8], [15], [22] can propose SQL or settings, but none checks proposals against an allow-list, demands measured evidence, separates reversible from irreversible actions, or requires human approval for the latter.
- **Security separated from performance.** Access-anomaly and injection detection [9], [24] run apart from tuning, so no component decides which action takes priority when they compete.
- **Evaluation in isolation.** Systems are tested on one task at a time; few are evaluated end to end on a live workload with injected faults and an ablation of their safety mechanisms.

Proposed Method

MADA separates *thinking* from *acting*. The agents never change production: they read telemetry, test ideas on a sandbox clone, and emit structured proposals — action SQL, rollback SQL, resources touched, risk level and evidence. Only the coordinator and safety gate apply changes. Figure 1 shows the architecture.

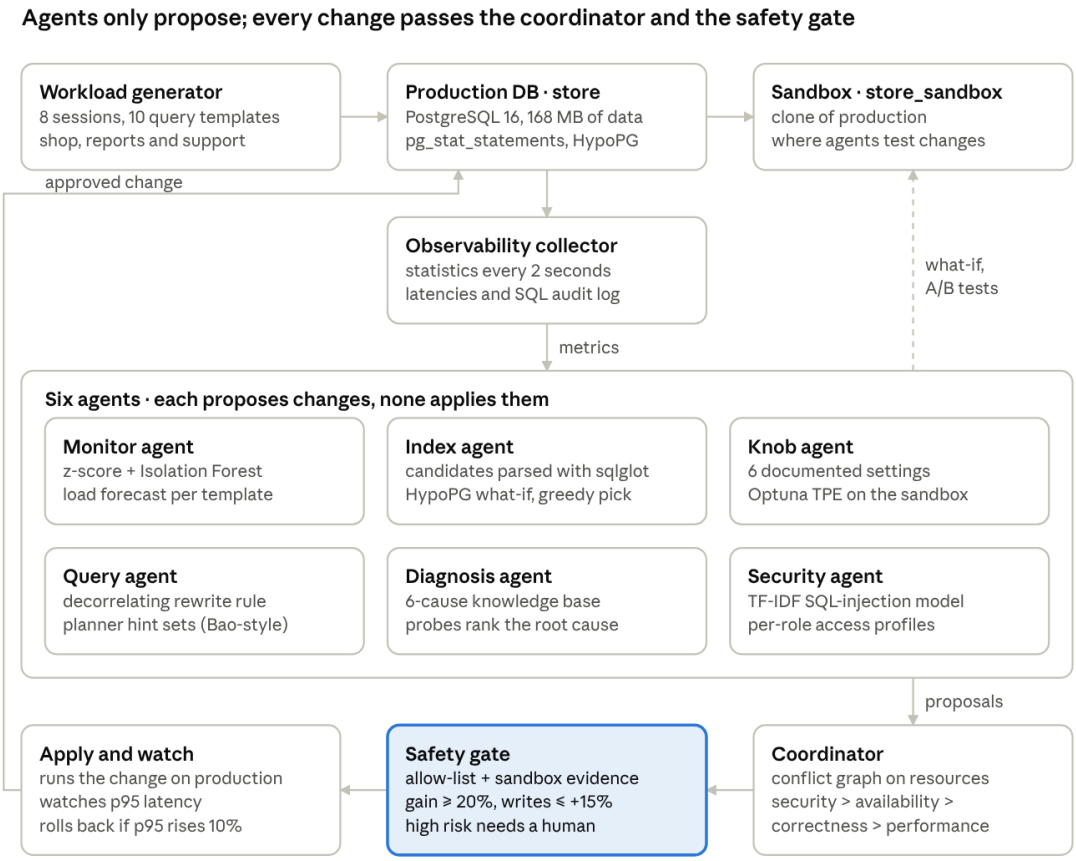


Figure 1. MADA architecture: telemetry in, proposals out, one gated path to production

Components

Testbed and workload generator. Builds an e-commerce schema of six tables (customers, products, orders, order_items, reviews, payments) with primary keys only, loads seeded data (168 MB), and clones it into a sandbox with CREATE DATABASE … TEMPLATE. Eight threads replay ten parameterised query templates under three roles (application, analyst, support). Every statement's latency and SQL go to a recorder that doubles as the audit log. Before each statement the workload reads two registry tables, mada.hints and mada.rewrites — the only way an agent can change the SQL the application runs.

Observability collector and monitor agent. Every 2 s the collector reads pg_stat_database, pg_stat_activity, pg_locks and pg_stat_user_tables, turns counters into per-second rates over a 10 s window, and adds host CPU and the 95th-percentile (p95) latency of each template. The monitor learns a baseline (mean and spread per metric plus an Isolation Forest [26] over all metrics) and raises an alert when a metric is more than 4σ from normal for two consecutive ticks and the forest agrees, or more than 12σ on its own. It also forecasts each template's arrival rate a minute ahead, following QB5000 [11].

Query agent. Takes analytical templates whose mean latency exceeds 250 ms. It tries rule-based rewrites first — rule R1 parses the SQL with sqlglot and turns a correlated COUNT(*) subquery into a LEFT JOIN on a grouped derived table, then checks that the rewrite returns identical rows for sample parameters — and then times Bao-style planner hint sets [6] on the sandbox. Only a variant at least 20% faster becomes a proposal.

Index agent. Reads the ten most expensive statements from pg_stat_statements, parses them with sqlglot to collect candidate one- and two-column indexes, and costs each candidate as a hypothetical index with HypoPG and EXPLAIN (GENERIC_PLAN) [16]. A greedy search in the style of Extend [17] keeps the candidates with the largest frequency-weighted cost reduction within a 120 MB budget. Because estimated cost is not run time [5], each pick is then built for real on the sandbox and the affected templates and the write path are timed with and without it.

Knob agent. Tunes six reload-only settings (work_mem, hash_mem_multiplier, random_page_cost, effective_cache_size, max_parallel_workers_per_gather, jit). A catalog written from the PostgreSQL manual gives each setting's meaning and safe range, as in GPTuner [15]; Optuna's TPE sampler [27], [28] runs 14 trials, each replaying the current workload mix on the sandbox weighted by observed frequency.

Diagnosis agent. Finds DBSherlock-style predicates [7] — metrics whose anomaly-window mean is at least 3σ from normal — and matches them against a knowledge base of six root causes (lock contention, work_mem spill, missing index, table bloat, connection storm, CPU saturation). Each cause has a metric signature and a confirming SQL probe; for lock contention the probe follows pg_blocking_pids() to the blocking session. The top cause's fix is routed to the agent that owns it.

Security agent. A TF-IDF model over character 2- to 5-grams with logistic regression, trained on 1,500 benign search statements and 700 injected ones from five attack families [24], screens every statement from the product search — the one place user input is spliced into SQL. An access monitor learns which tables each role touches in normal operation [9] and flags reads outside that profile; the response revokes the grant and ends the role's sessions.

Coordinator and safety gate. Every proposal declares the resources it touches (a table, the planner, memory, a session or a role). The coordinator orders proposals by priority (security > availability > correctness > performance), then by gain and risk, and defers any proposal whose resources are already claimed in the current window. The safety gate then checks the rules in Table 2. After applying a change, the coordinator watches the p95 latency of the affected templates and rolls the change back if it rises by more than 10%.

Table 2. Safety-gate rules, checked in order

Check	Rule
Static policy	Every statement matches an allow-list of statement shapes (CREATE/DROP INDEX, ALTER SYSTEM for catalogued knobs, registry inserts, VACUUM, GRANT/REVOKE, pg_terminate_backend); settings lie in their documented ranges; reversible actions carry a rollback statement.

Check	Rule
Sandbox evidence	Performance proposals show at least 20% measured gain on the sandbox and at most 15% extra write cost; rewrites return identical rows on sample parameters.
Risk	Low and medium risk apply automatically; high-risk, irreversible actions (ending sessions, revoking access) need human approval.
Watch and rollback	After apply, p95 of the affected templates is watched; a rise of more than 10% triggers the rollback statement.

Workflow

Tuning cycle. (1) The workload runs and the collector records two minutes of baseline. (2) The query agent fixes slow statements first, so later agents tune the SQL that production will actually run. (3) The index agent proposes indexes from a fresh minute of statistics. (4) The knob agent searches configurations against the tuned workload. (5) After each agent, the coordinator runs one or more windows: order, defer conflicts, gate, apply, watch, roll back if needed.

Incident loop. Every 2 s the monitor checks the latest metrics and the security agent scans new audit entries. A security finding goes straight to the coordinator with top priority. A performance alert triggers the diagnosis agent, which ranks causes, confirms the top one with its probe and proposes the owning agent's fix; the coordinator and gate then decide as above.

Implementation

The prototype is about 2,200 lines of Python in 13 modules, running against PostgreSQL 16 with the pg_stat_statements and HypoPG extensions. It uses psycopg 3 for database access, sqlglot for SQL parsing and rewriting, scikit-learn for the Isolation Forest and the injection classifier, and Optuna for Bayesian optimisation. A live web dashboard shows the agents' log, every coordinator and safety-gate decision, and the latency of each query template, and lets the operator inject faults and approve or reject high-risk actions. LLM support is optional: the prototype runs in rule mode, with explanations taken from the knowledge base.

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